Abhishek Jha

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Education

Indraprastha Institute of Information Technology, Delhi Nov 2022 – May 2027
B.Tech in Electronics and VLSI *Delhi, India*

- **Relevant Coursework:** Artificial Intelligence, Algorithm Design & Analysis, Data Structures & Algorithms, Probability & Statistics, Operating Systems, Computer Organization

Technical Skills

ML & Foundation Models: PyTorch, TensorFlow, JAX, Transformers, Hugging Face, LangChain, LLM Evaluation
RL & Agentic Systems: PPO, Monte Carlo Tree Search, OpenEnv, WebMCP / MCP, Long-Horizon Agents
Languages: Python, C++, C, Verilog
Systems: Linux, Git, Docker, FastAPI, GCP, Qualcomm AI Hub, TFLite, Bazel, Amazon Neptune/Gremlin

Experience

Amazon Jan 2026 – Jun 2026
Applied Science Intern *Agentic AI & ML Systems*

- Designed scalable memory services for long-horizon agentic systems, keeping retrieval **flat at ∼3.8K tokens** regardless of history length across multi-step workflows.
- Built a **dual-graph memory architecture** (episodic events + evidence-scored insight graph) with cross-user knowledge transfer, outperforming Mem0, ZEP, and MAGMA baselines by up to **1.48** points (LongMemEval) and **0.93** points (LoCoMo QA); converted to a **Pre-Placement Offer, Applied Scientist**.

STMicroelectronics Jun 2025 – Dec 2025
Deep Learning Quantization Intern *Python, PyTorch, CNNs*

- Quantized **YOLOv5s** to 5-bit (**6.4**× compression) via hardware-friendly Vector Quantization and Sparse Error Compensation, with only a **3.3%** relative accuracy drop; used Qualcomm AI Hub to quantize OpenAI CLIP to int8 precision.
- Applied **Channel Relaxed Vector Quantization (CRVQ)** to a **Vision Transformer (ViT)**, achieving **5.2-bit** (**6.12**× compression) via Adam-based codebook fine-tuning, with only a **0.83%** relative accuracy drop.
- Wrote a custom int8 **SiLU** layer as a **TensorFlow** custom op (built with **Bazel**) for an IBM × ST **TFLite** activation-testing project, cutting activation-approximation overhead by **40%**.

Centre for Intelligent Product Development (CiPD), IIIT Delhi May 2024 – May 2025
Software and Product Development Intern *Delhi, India*

- Coordinated cross-functional software and hardware teams to design custom **STM32** expansion boards and a full-stack IoT system, achieving **99.3%** accuracy on real-time sensor data processing across **780K+** datapoints. GitHub

Projects

Agent Memory Bus: Cross-Session Memory for Browser Agents | *JavaScript, WebMCP* 2026

- Built a client-side dual-graph memory system (episodic observations + semantic concept graph) exposing 6 agent-callable tools over WebMCP with local embeddings and zero server/API cost; hardened against stored indirect prompt injection under a documented threat model.

DataLineageEnv: Long-Horizon RL Environment for Data-Lineage Agents | *Python, PPO, OpenEnv* 2026

- Built an OpenEnv-compliant long-horizon simulation environment with deterministic programmatic verification for reproducible agent evaluation; shipped a PPO training pipeline and multi-model inference harness backed by a 57-test suite at 95% coverage.

Buy or Wait?: A Deterministic Financial-Safety Agent | *Python* 2026

- Built a financial-affordability agent combining narrow LLM extraction with an auditable, deterministic decision engine; validated by fault injection (10 deliberately malformed rows, each caught) and byte-for-byte reproducible output.

Publications & Research Output

ORCID: 0009-0005-9169-5606 ·Google Scholar: Profile

- **Verified Confinement: A Case-Study Pattern for Confining LLM Output Behind Deterministic Verifiers**, *NeurIPS 2026 WS* – Accepted.
- **QUARTZ: Dequantization-Free LLM Attention via Lookup Tables**, *COLM 2026 WS* – Published & presented; **7.4**× read-bandwidth reduction vs. FP16.
- **STARK: Moving Towards Agents That Truly Learn and Remember**, *AAAI-27* – Accepted.
- **CIM-FLEX: An Integer-Only Flexible Periphery for Distributed Compute-In-Memory Architectures**, *ISLPED 2026* – Published (STMicroelectronics, IBM, EPFL).

Achievements

- **Pre-Placement Offer (PPO)**, Applied Scientist, Amazon.
- **All India Rank 3**, Amazon ML Challenge 2025 (20,000+ teams); **Init-Together Award**, Amazon 2026.
- **All India Rank 2**, GE HealthCare Precision Care Challenge 2026 Grand Finale (8,000+ teams, team of 4) – ClaimCast.
- **Two Research Posters Accepted**, Amazon internal EdgeAI Science Summit (Feb 2026).